PSYC1071: Introduction to Psychology 1
Animal Learning and Cognition

Lecture 2:
Skinner and Pavlov

Professor Anthony McGregor

Durham University
Department of Psychology

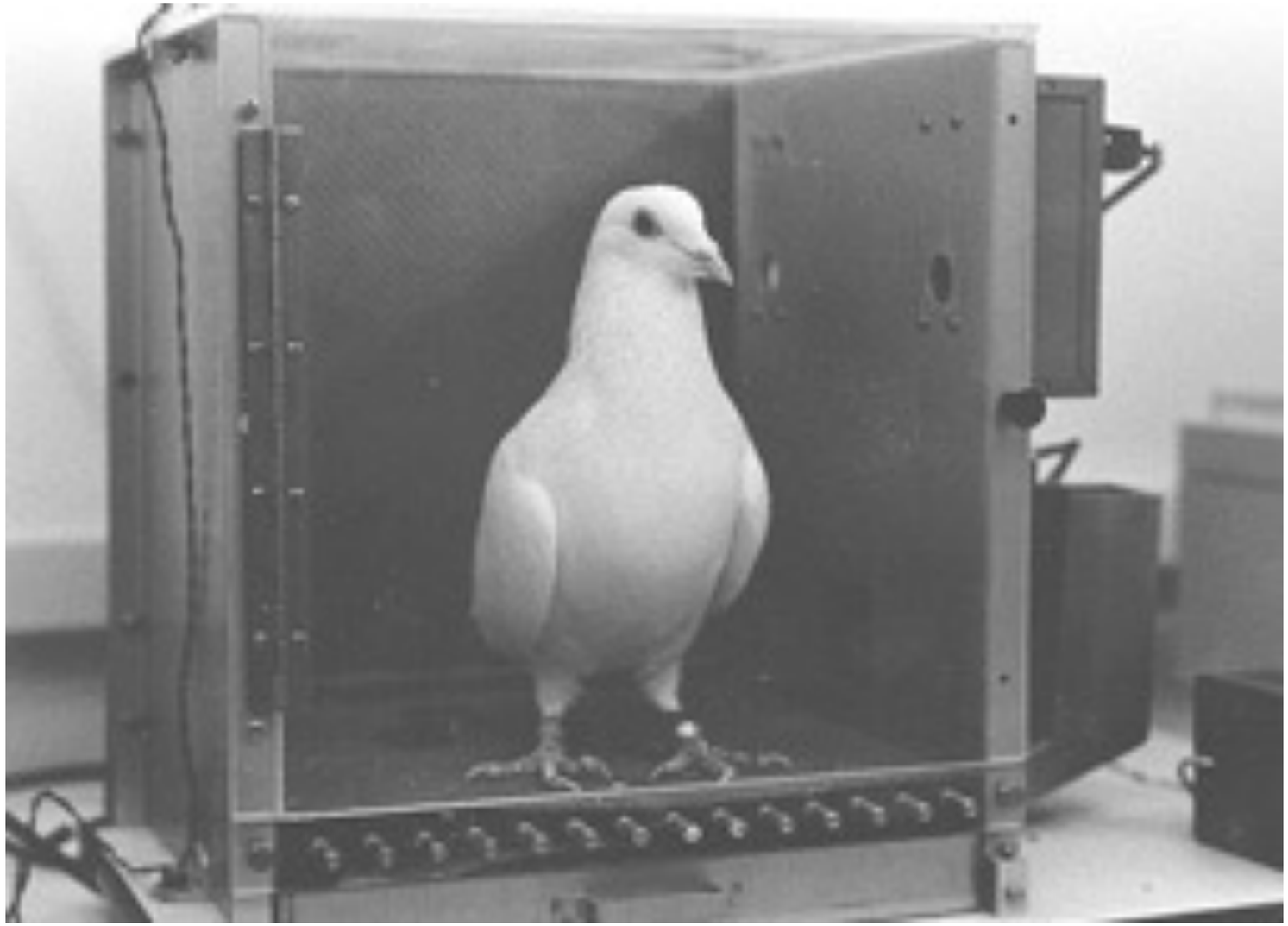
Historical Background

- Charles Darwin (1809-1882)
- Edward Thorndike (1874-1949)
- John B Watson (1878-1958)
- Burrhus Frederick (BF) Skinner (1904-1990)
- Ivan P Pavlov (1849-1936)

"The major problems of the world today can be solved only if we improve our understanding of human behavior" *About Behaviorism (1974)*

Burrhus Frederick Skinner

1904 - 1990

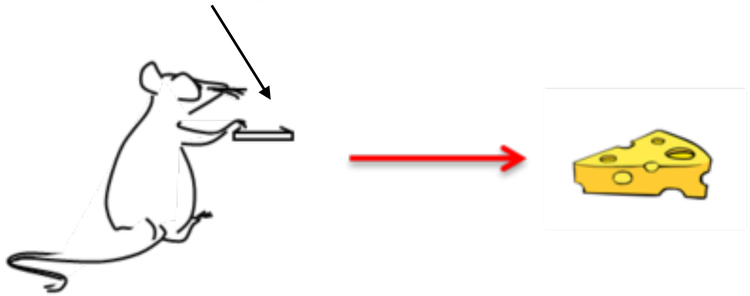


Historical Background

Instrumental conditioning

(Operant)

(Stimulus)



Response

Outcome
(reinforcer)

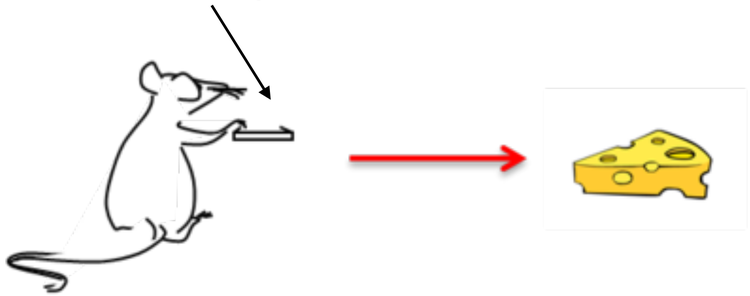
- Operant or Instrumental conditioning
- Response shaping
- Schedules of reinforcement

Historical Background

Instrumental conditioning

(Operant)

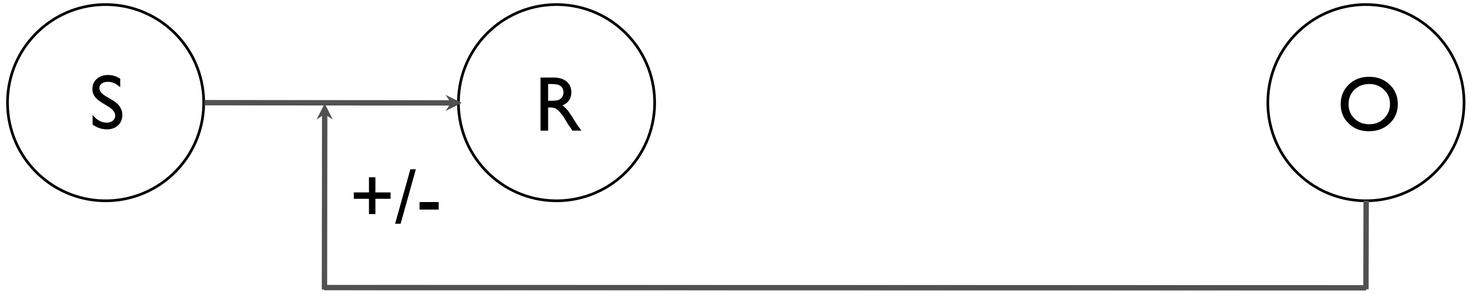
(Stimulus)



Response

Outcome
(reinforcer)

Thorndike's Law of Effect



If a response leads to a satisfying outcome
it will be strengthened

The outcome (O) only strengthens or weakens the connection between S and R, and the outcome itself is not represented in animals on making the instrumental response.

Do animals only learn about a response?

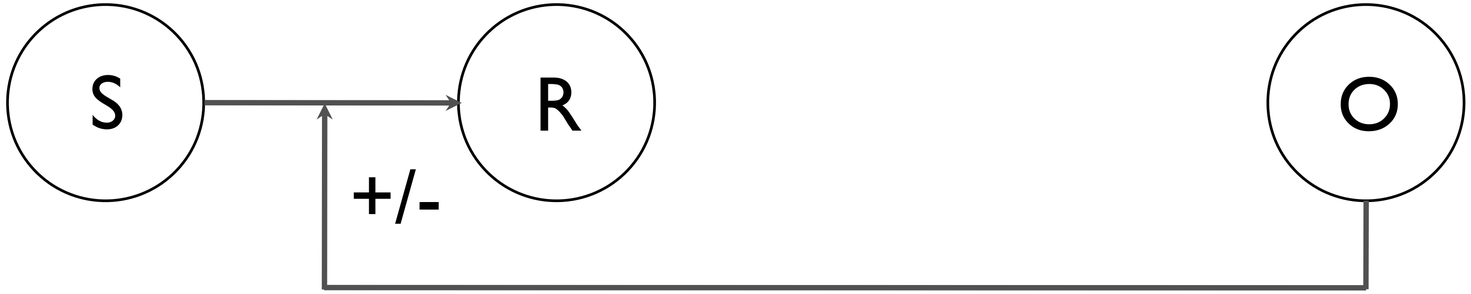
Colwill and Rescorla (1985)

Do animals only learn about a response?

Colwill and Rescorla (1985)

Stage 1	Stage 2	Test
Lever press → Food and Chain pull → sucrose	Food → Illness and Sucrose → nothing	Lever and Chain

Thorndike's Law of Effect



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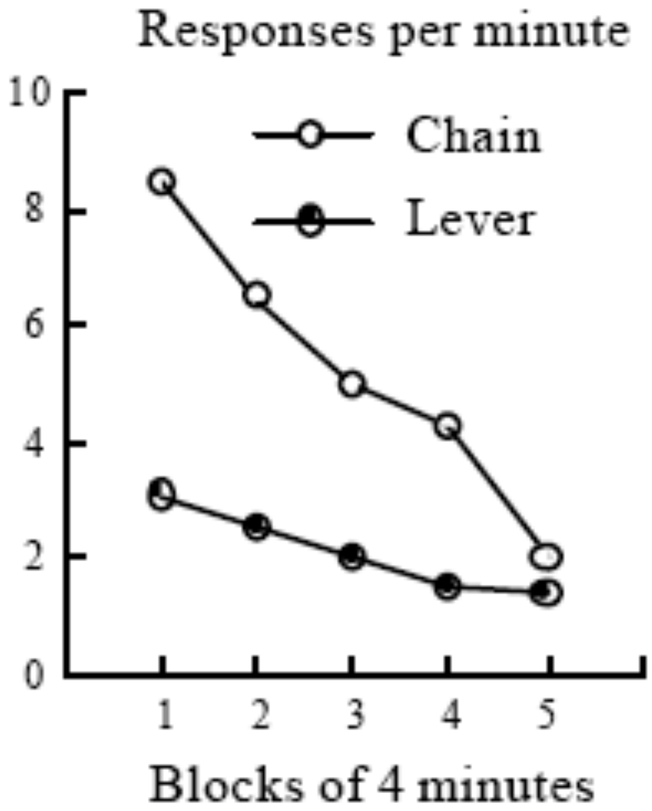
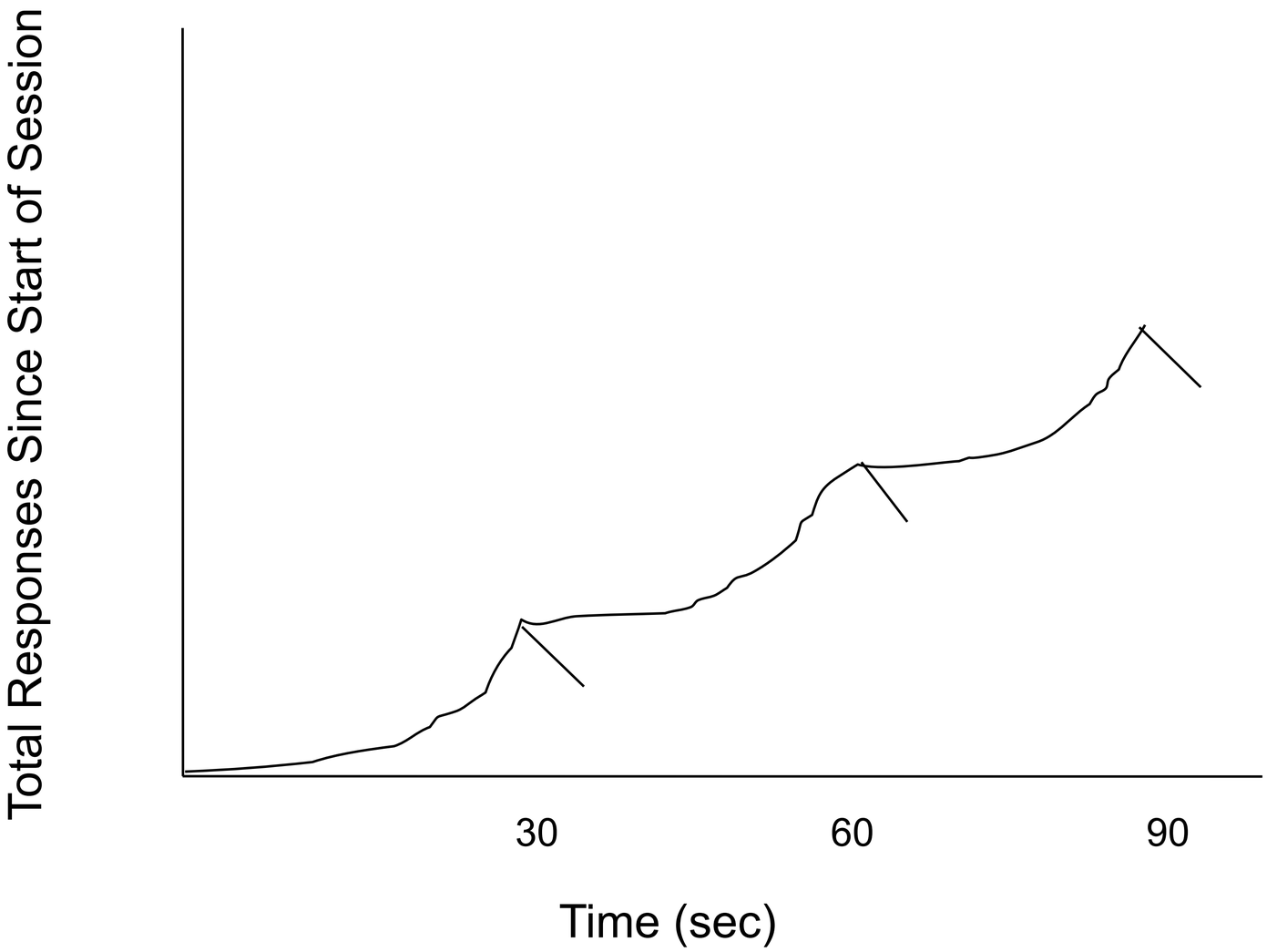


- Operant or Instrumental conditioning
- Response shaping
- Schedules of reinforcement

Cumulative Record for a Fixed Interval (FI) Schedule

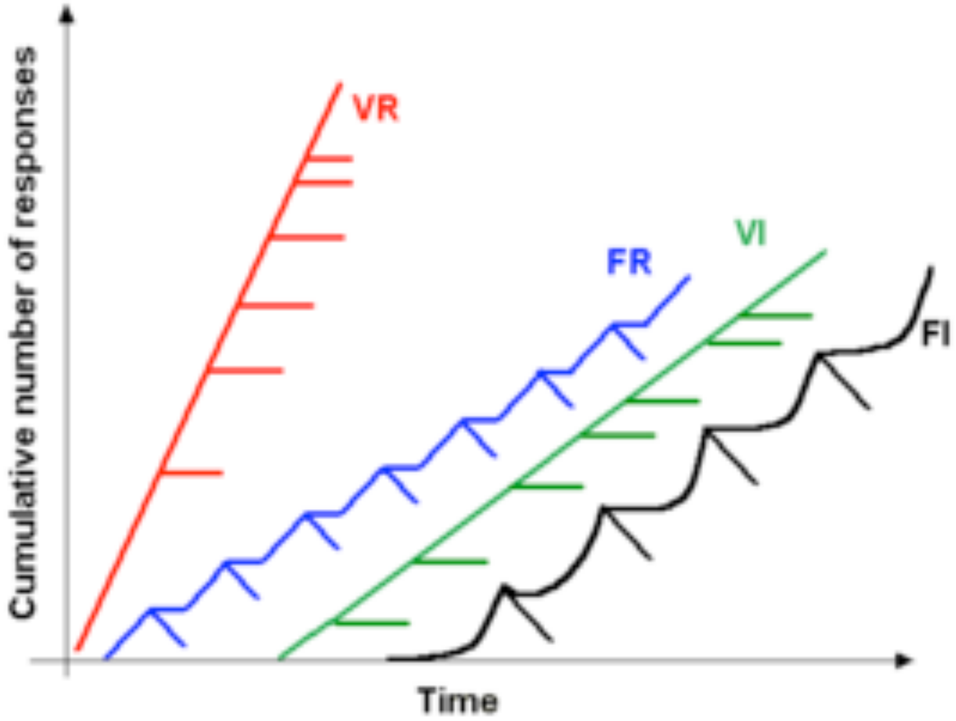


Schedules of reinforcement

	Ratio	Interval
Fixed	FR	FI
Variable	VR	VI

Schedules of reinforcement

	Ratio	Interval
Fixed	FR	FI
Variable	VR	VI



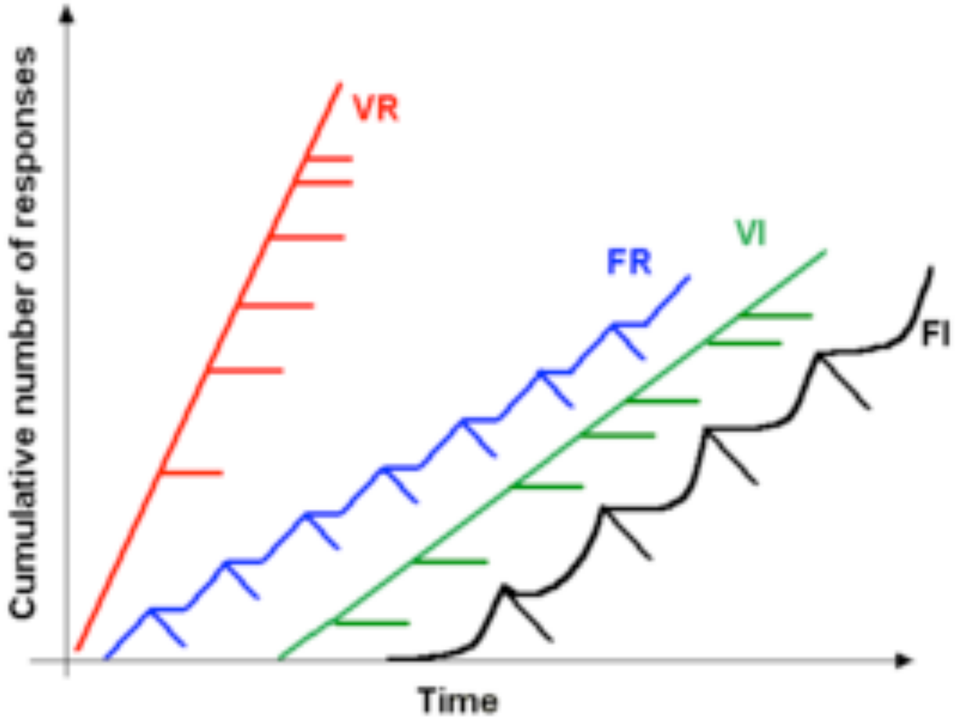
Schedules of reinforcement

FI - studying for exams

FR - ‘piece work’ in factories (gig ecomony?)

VI - instagram

VR - gambling



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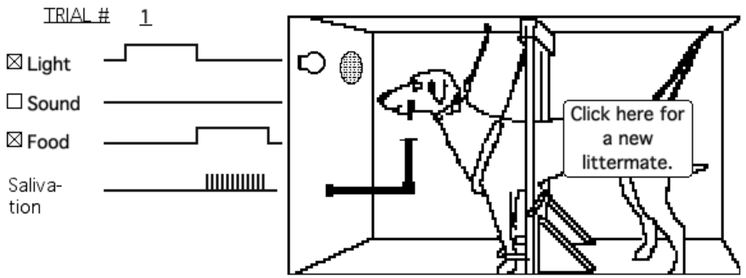
Pavlovian or Classical conditioning

Fig. 5.2. Sara and Ivan Pavlov at about the time of their wedding

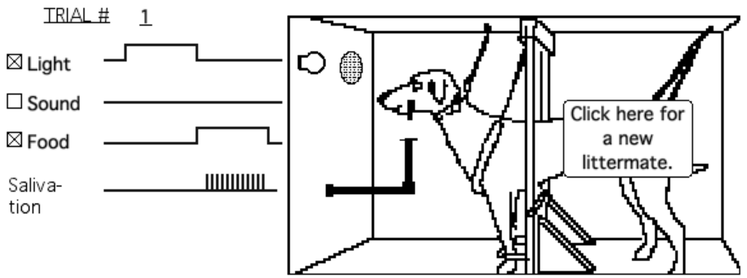
Pavlovian conditioning



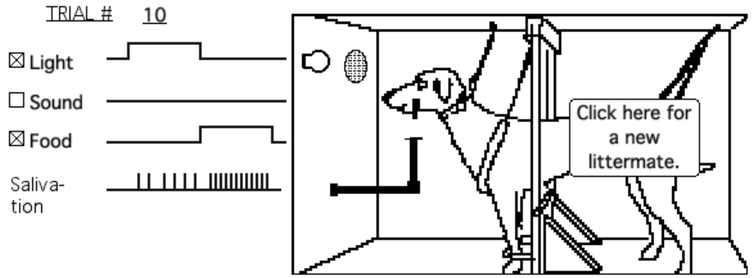
CS → US
↓
UR

CS = conditioned stimulus (light)
US = unconditioned stimulus (food)
UR = unconditioned response (salivation)
CR = conditioned response (salivation)

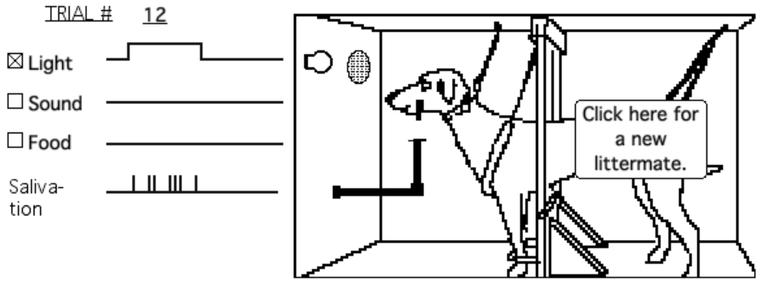
Pavlovian conditioning



CS → US
↓
UR



CS → US
↓ ↓
CR UR

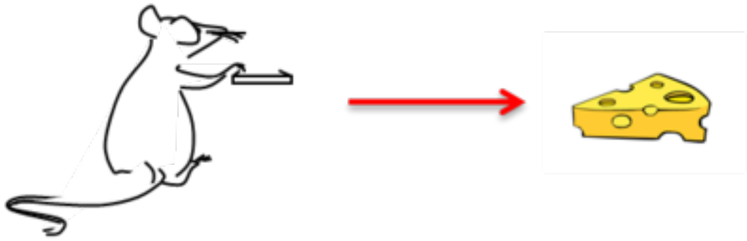


CS
↓
CR

Historical Background

Instrumental conditioning

(Operant)



Response

Outcome
(reinforcer)

Pavlovian conditioning

(Classical)

Stimulus
(CS)

Stimulus
(US)

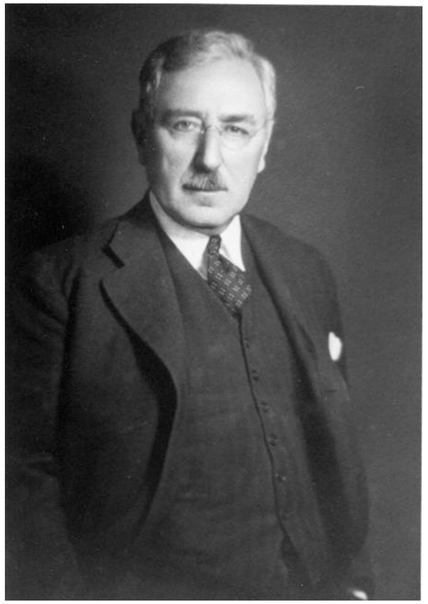
Before conditioning

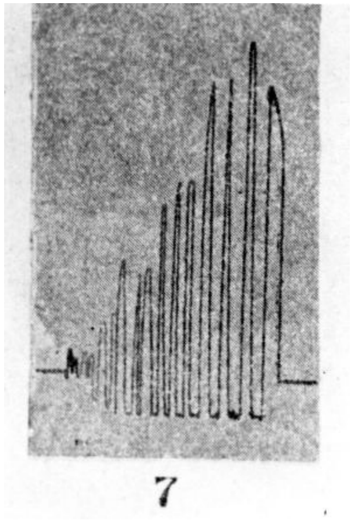
Bell	→	No Response
Food (unconditioned stimulus, US)	→	Salivation (unconditioned response, UR)

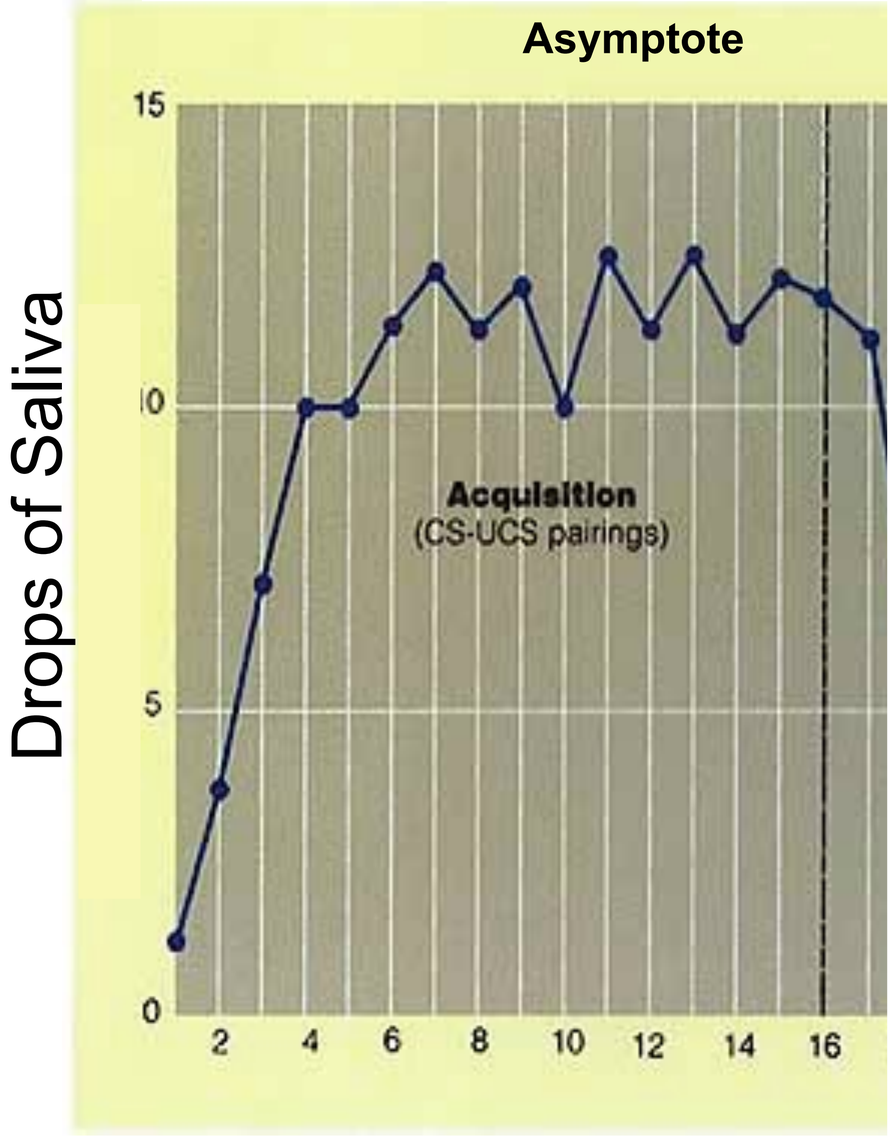
During conditioning

Bell	→	Food (unconditioned stimulus, US)
Bell (conditioned stimulus, CS)	→	Salivation (conditioned response, CR)

Edwin B. Twitmyer

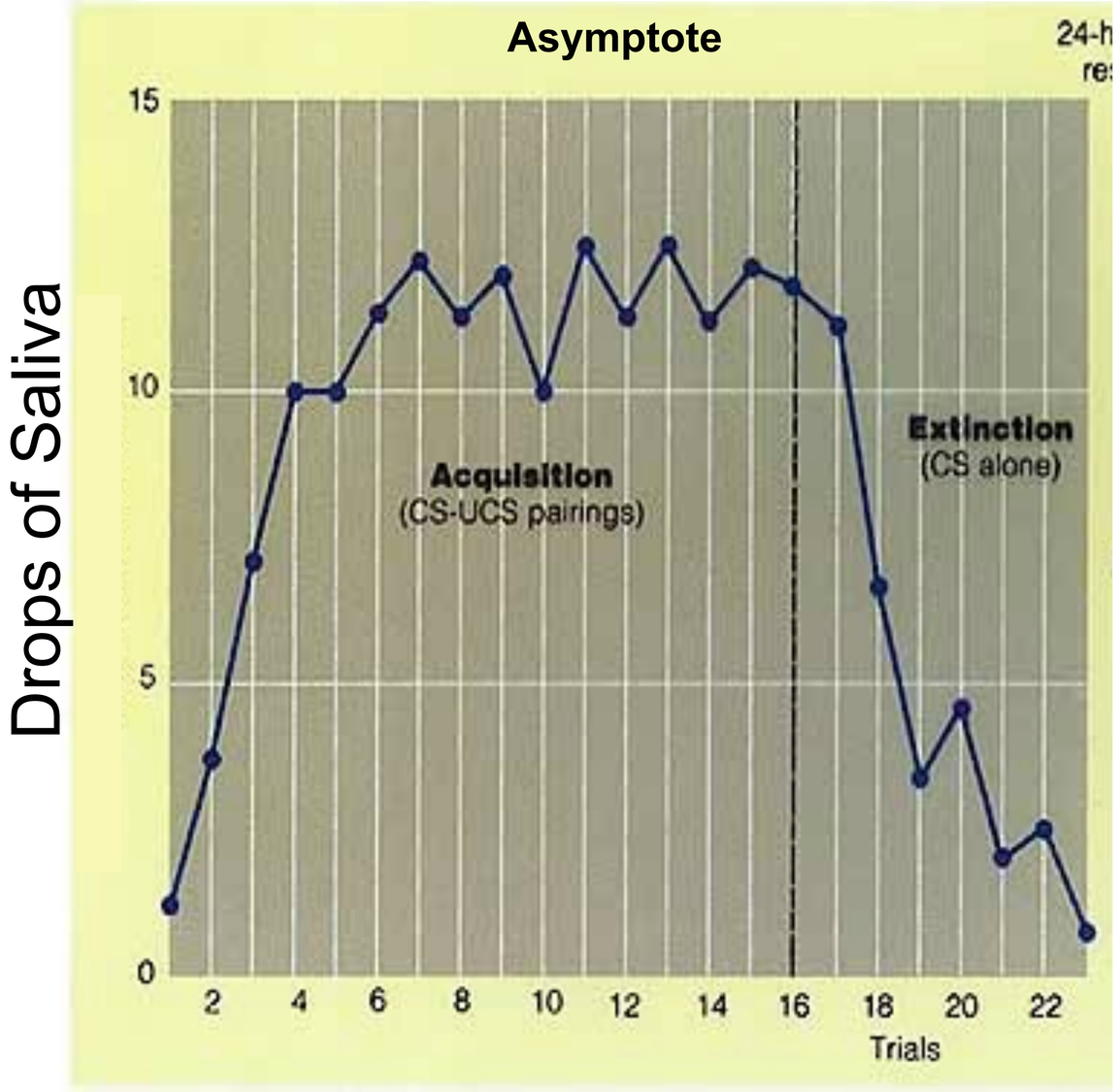






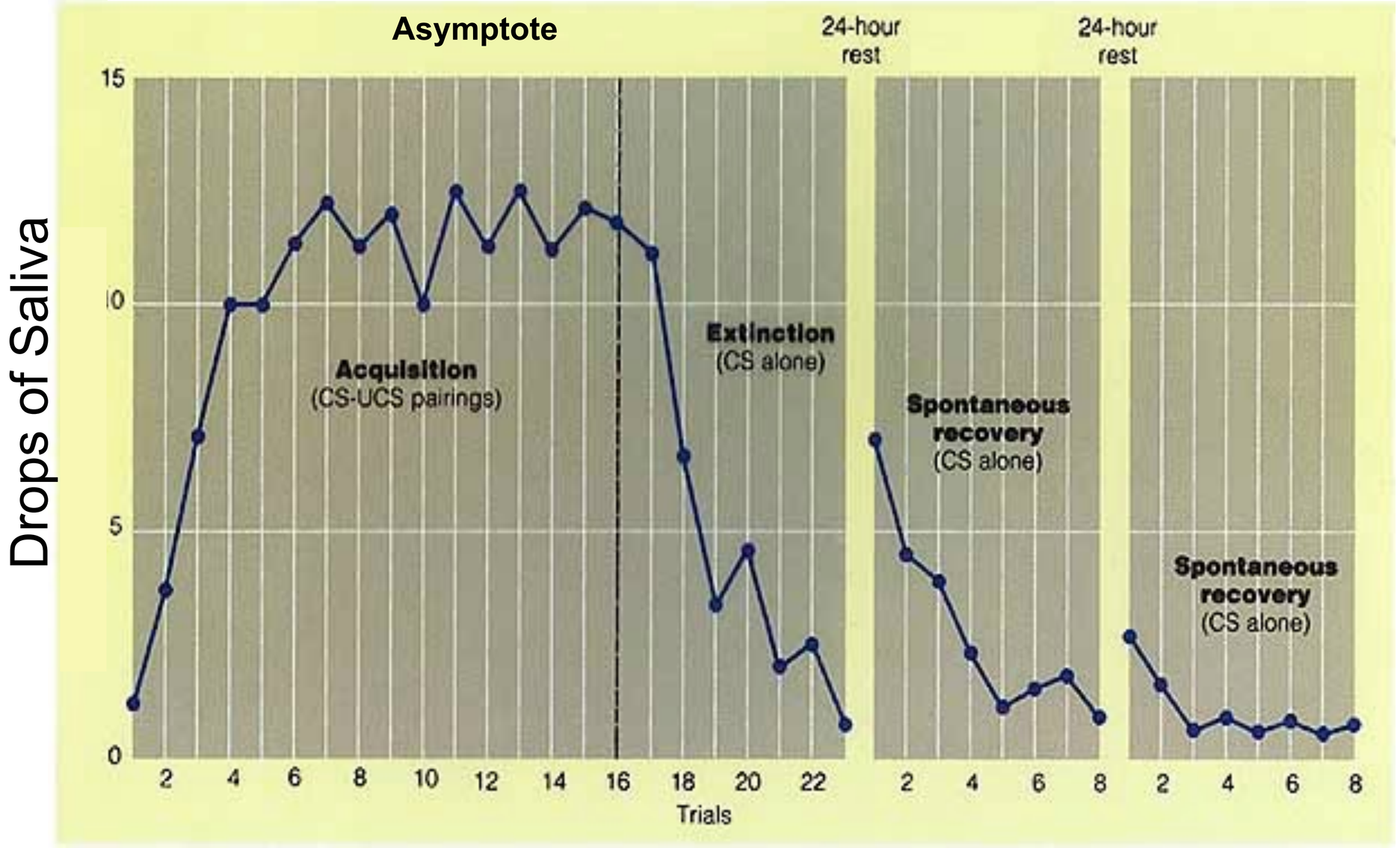
Acquisition

CS-US pairings



Acquisition
CS-US pairings

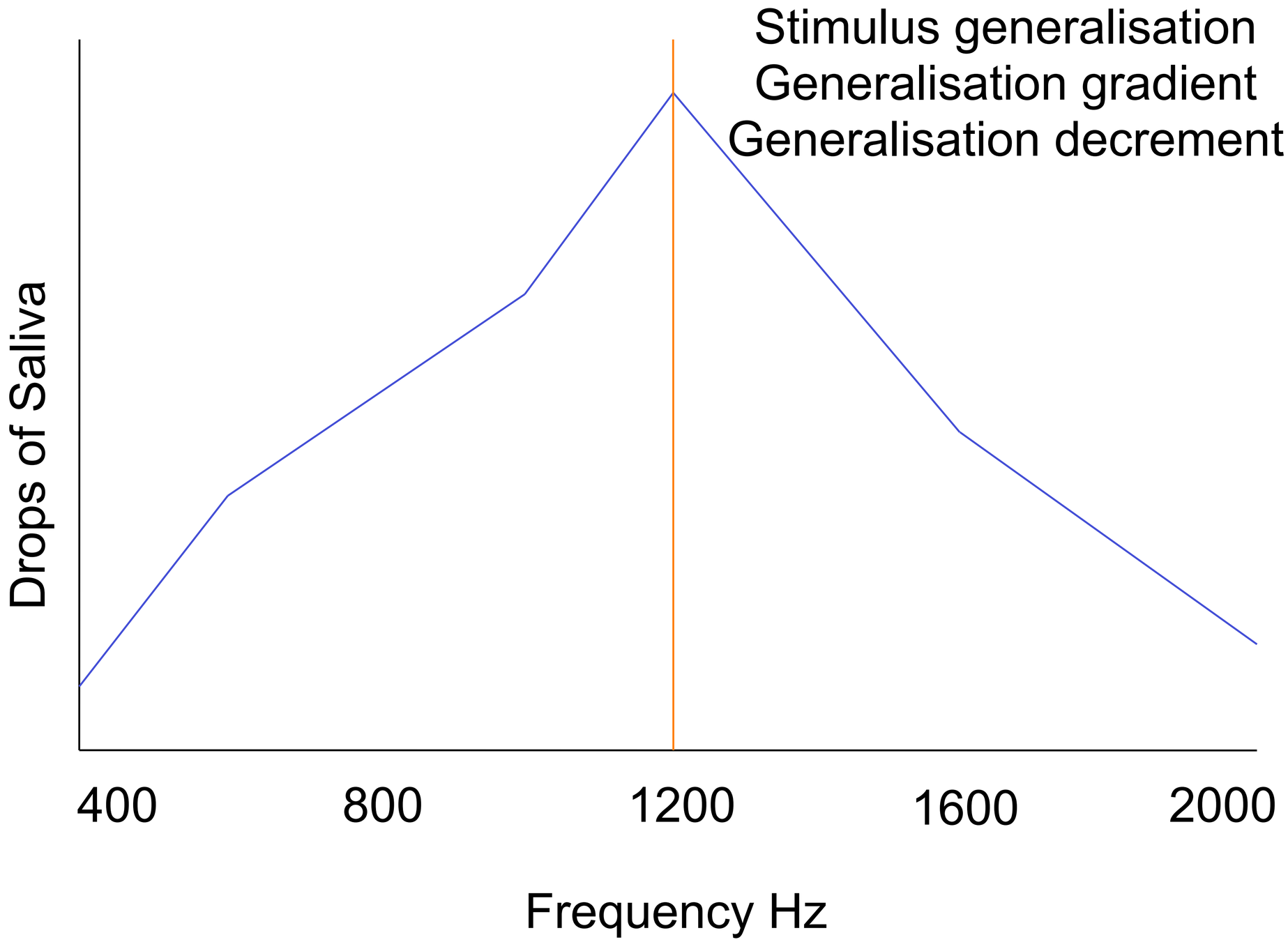
Extinction
CS alone



Acquisition
CS-US pairings

Extinction
CS alone

Spontaneous Recovery
CS alone


Stimulus generalisation
Generalisation gradient
Generalisation decrement
Drops of Saliva
400
800
1200
1600
2000
Frequency Hz

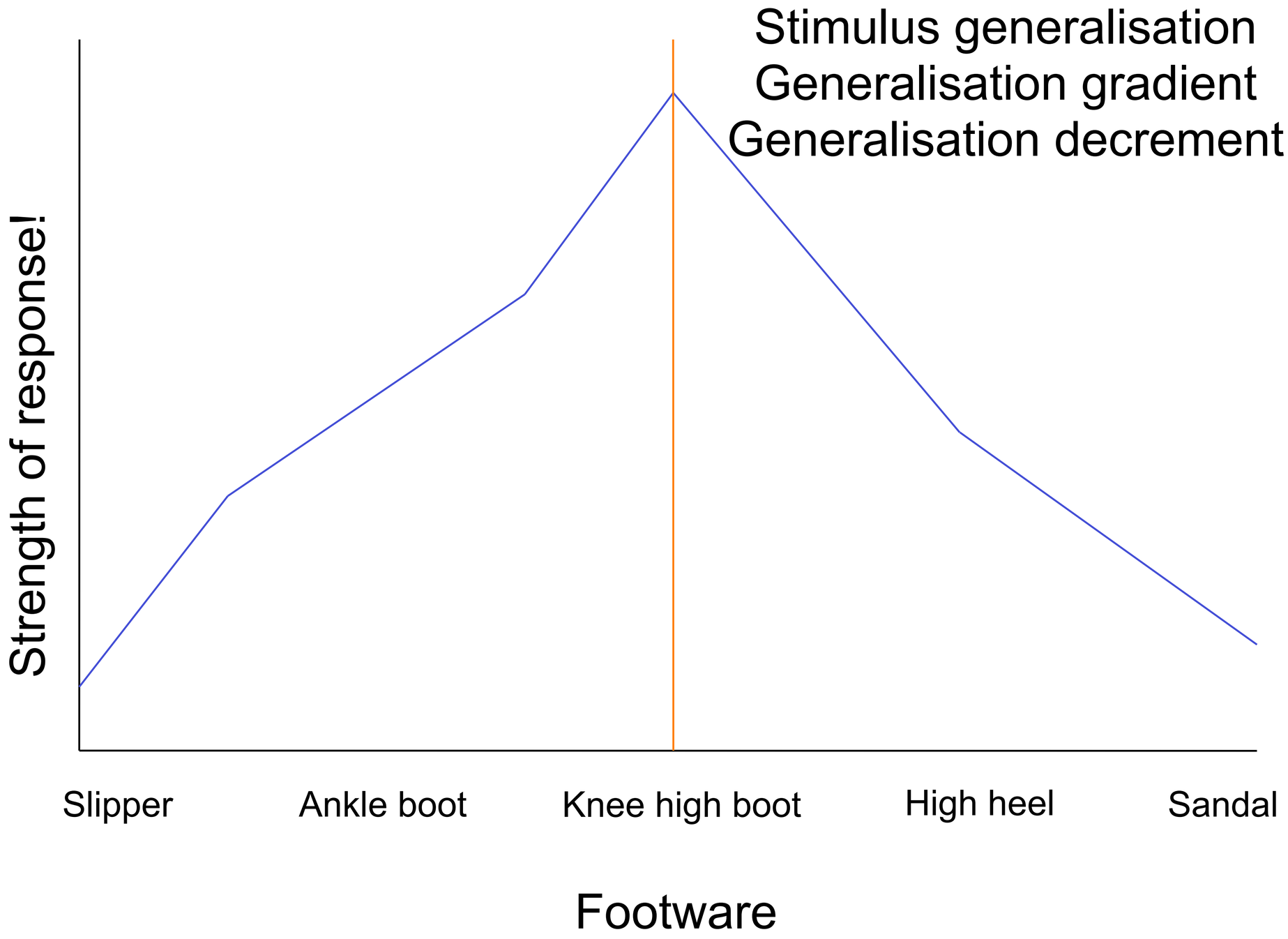
Rachman, S J (1966) Psychological Record, 16, 293-296

Before conditioning

Leather Boot	→	No Response
Picture (unconditioned stimulus, US)	→	?? (unconditioned response, UR)

During conditioning

Leather Boot	→	Picture (unconditioned stimulus, US)
Leather Boot (conditioned stimulus, CS)	→	?? (conditioned response, CR)


Stimulus generalisation
Generalisation gradient
Generalisation decrement
Strength of response!
Slipper
Ankle boot
Knee high boot
High heel
Sandal
Footware

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Next lecture - the study of animal intelligence